

**Indivisible beable matter as a source for unimodular semiclassical
gravity: gravitational decoherence, a dynamical cosmological
term, and the covariance obstruction**

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Abstract

A central obstruction to coupling classical gravity to quantum (beable) matter is that the contracted Bianchi identity forces covariant conservation of the source, $\nabla^\mu T_{\mu\nu} = 0$, whereas a stochastic, indivisible beable process injects energy–momentum and so violates it. We show that *unimodular gravity* removes this obstruction: its trace-free field equations do not impose conservation, and the non-conservation is absorbed into a dynamically generated cosmological term (Josset–Perez–Sudarsky). Using the indivisible-stochastic-process (ISP) beable of Barandes as the matter source, the violation current $\eta_\nu = \nabla^\mu T_{\mu\nu}$ decomposes into a *mean* part and a *fluctuating* part. The mean part, homogeneous and isotropic in the cosmic rest frame, is curl-free and sources a dynamical cosmological term $\Lambda(t)$; we find, however, that for Diósi–Penrose-type sourcing this term is quantitatively *excluded* as dark energy — the same regularization length the spontaneous-radiation experiments bound caps the induced Λ many orders below the observed value. The fluctuating part sources metric noise (an Einstein–Langevin term) that, in the Newtonian limit, reduces to the Diósi–Penrose / Tilloy–Diósi gravitational decoherence and to the Gaussian-onset, falsifiable signature of a companion paper. We resolve the recent dispute over the entanglement signature for this construction — the beable-sourced field is a classical channel and so predicts a *negative* Bose–Marletto–Vedral result (falsifiable), with mandatory complementary decoherence — and we isolate the covariance of the beable dynamics for a quantum field as the single binding open problem, dissecting it down to a four-way no-free-lunch among manifest covariance, real time, genuine probability, and a physical-time-evolving beable.

I. INTRODUCTION AND THE THREE COUPLINGS

How a classical spacetime couples to quantum matter admits three structurally distinct answers, experimentally distinguishable in principle [30–32]: (1) *quantized gravity* — the metric is itself in superposition and can mediate entanglement (BMV-positive); (2) *mean-field semiclassical gravity* (Schrödinger–Newton), $G_{\mu\nu} = 8\pi G\langle T_{\mu\nu} \rangle$, which is pathological (superluminal signalling, an experimentally constrained nonlinear self-gravity); (3) *stochastic semiclassical gravity* — the field is classical but sourced by the *actual localized* matter, consistent in the Newtonian limit and yielding gravitational decoherence as a by-product [6].

The indivisible-stochastic-process (ISP) reformulation of quantum theory [1, 2] is a natural home for option 3: its primitive ontology is a definite configuration (beable) at all times, exactly the localized source option 3 needs and that Tilloy–Diósi supplied by hand through collapse outcomes. This paper asks the relativistic question: *can the ISP beable source classical gravity covariantly?* The immediate obstruction is energy conservation; the main result is that unimodular gravity removes it, at the price of a dynamical cosmological term and a residual covariance problem we then isolate sharply.

Scope note. The gravity-sourcing results (Results 1–6) use only that the matter is a *non-conserved source*; they are **Markovian-agnostic** and hold for the ISP beable treated as a non-relativistic / schematic source, exactly as for ordinary collapse matter (the original setting of Josset–Perez–Sudarsky). “Indivisible” names the program’s ontology, not a load-bearing property of Results 1–6; wherever “indivisible” or “relativistic ISP” appears as a fully covariant interacting-field beable process, that object is the *open* problem of Sec. X, not an input to the sourcing results. The non-Markovian content bites in exactly two places: the decoherence onset shape (the Gaussian signature) and the covariant-construction lever.

Claims are marked **[STD]** (established), **[SYN]** (this synthesis), or **[OPEN]**. Nothing is asserted as a solution of quantum gravity or an experimentally confirmed deviation from general relativity.

II. THE OBSTRUCTION: BIANCHI VERSUS BEABLE NON-CONSERVATION

In general relativity, $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ with the contracted Bianchi identity $\nabla^\mu G_{\mu\nu} \equiv 0$ forces $\nabla^\mu T_{\mu\nu} = 0$. An ISP beable process injects energy–momentum (record-stabilization, the indivisible analogue of collapse, makes the stress–energy built from the actual configuration jump and diffuse), so that

$$\nabla^\mu T_{\mu\nu} = \eta_\nu \neq 0, \tag{1}$$

with η_ν a stochastic violation current. The left side of Einstein’s equation is identically divergence-free; the right side is not. The naive coupling is inconsistent — the obstruction that has made relativistic collapse / beable gravity difficult [33]. **[STD]**

III. UNIMODULAR GRAVITY REMOVES THE OBSTRUCTION

Unimodular gravity (UG) fixes $\sqrt{-g}$ and yields the *trace-free* Einstein equations [8, 9]:

$$R_{\mu\nu} - \frac{1}{4}g_{\mu\nu}R = 8\pi G(T_{\mu\nu} - \frac{1}{4}g_{\mu\nu}T). \quad (2)$$

Conservation is no longer imposed. Taking the divergence and using $\nabla^\mu R_{\mu\nu} = \frac{1}{2}\nabla_\nu R$,

$$\frac{1}{4}\nabla_\nu(R + 8\pi G T) = 8\pi G \eta_\nu. \quad (3)$$

Defining the scalar Λ via $R + 8\pi G T \equiv 4\Lambda$, two cases follow:

$$\begin{aligned} \eta_\nu = 0 : \Lambda &= \text{const (usual UG integration constant),} \\ \eta_\nu \neq 0 : \nabla_\nu \Lambda &= 8\pi G \eta_\nu \text{ (}\Lambda \text{ dynamical, integrating the violation).} \end{aligned}$$

(4)

So in UG the non-conservation is absorbed into a dynamical cosmological term; consistency needs only that η_ν be a gradient ($\eta_\nu = \nabla_\nu \phi$, so $\Lambda = 8\pi G \phi + \Lambda_0$). This is the mechanism of Josset–Perez–Sudarsky [7], with the ISP beable a concrete realization. [**STD mechanism; SYN applied to ISP**]

Result 1 [SYN]. The conservation-vs-Bianchi obstruction to sourcing relativistic gravity from *any* non-conserved matter (the ISP beable being one instance) is not fatal. Unimodular gravity evades it, converting the violation into a dynamical cosmological term, provided the violation current is curl-free. The mechanism uses only non-conservation — it is **Markovian-agnostic**, holding equally for ordinary collapse matter (the original setting of [7]); indivisibility is not used here.

IV. THE MEAN/FLUCTUATION SPLIT

Decompose $\eta_\nu = \langle \eta_\nu \rangle + \delta\eta_\nu$.

Mean — a *dynamical cosmological term*. For a record-stabilization process homogeneous and isotropic in the cosmic rest frame, $\langle \eta_i \rangle = 0$ while $\langle \eta_0 \rangle = w(t)$ is a net energy-density injection rate. Then $\langle \eta_\nu \rangle = (w(t), \mathbf{0})$ is curl-free, $\langle \eta_\nu \rangle = \nabla_\nu \phi$ with $\phi(t) = \int w dt$, and

$$\Lambda(t) = \Lambda_0 + 8\pi G \int^t w(t') dt'. \quad (5)$$

This is the Josset–Perez–Sudarsky structure, here sourced by the ISP rate. [**SYN**]

Fluctuation — gravitational decoherence. The fluctuating $\delta\eta_\nu$ is not curl-free; it sources metric fluctuations through the trace-free equations — an Einstein–Langevin noise term [10] — dephasing a massive system in spatial superposition. [SYN]

Result 2 [SYN]. The single ISP non-conservation splits into a mean part (curl-free, cosmic rest frame) sourcing a dynamical cosmological term, and a fluctuating part sourcing metric noise and hence gravitational decoherence. Dark-energy-like behaviour and gravitational decoherence are two faces of the same beable non-conservation.

V. THE NEWTONIAN LIMIT AND THE ENTANGLEMENT QUESTION

In the weak-field, non-relativistic limit UG reduces to $\nabla^2\Phi = 4\pi G\rho$, sourced by the beable mass density $\rho \geq 0$ (no conservation constraint: there is no Bianchi identity in the Newtonian limit). The fluctuating source reproduces, term for term, the Tilloy–Diósi beable-sourced gravity [6]: the Newtonian pair potential up to a short-distance cutoff plus a gravitational decoherence term. With the indivisible (non-divisible) survival law, the decoherence is non-exponential, with the falsifiable Gaussian onset

$$C(T) = C(0) \exp\left[-\frac{1}{2}(T/\tau_G)^2\right], \quad \tau_G = \hbar/E_G, \quad (6)$$

E_G the Diósi–Penrose self-energy of the mass-density difference [4, 5]. [SYN; reduces to STD]

The entanglement signature — resolved for the ISP construction. The ISP-beable-sourced field is, by construction, a *classical channel*: gravity is sourced by the definite beable (one value per mass), not the superposed wavefunction. Kafri, Taylor and Milburn [11] proved that a classically-mediated gravitational interaction — which they show is *equivalent to Diósi’s model* — cannot create entanglement, while necessarily producing decoherence. Entanglement cannot increase under local operations and classical communication, and the beable-sourced field carries only classical information about the actual configuration.

Result 4 [SYN]. The ISP-beable-sourced construction predicts a *negative* BMV result (no gravitationally-induced entanglement) together with mandatory, complementary gravitational decoherence, the two linked by the classical-channel structure [11]. A *positive* BMV result would falsify the construction.

This resolves the 2025 dispute as a model-definition issue. Analyses finding entanglement [12] treat gravity as a quantum two-body potential on superposed positions (with Diósi–Penrose decoherence on top); those finding none [11, 13] treat gravity as a classical channel sourced by the localized matter. The ISP ontology falls unambiguously on the classical-channel side. (Such a channel can still generate quantum *discord* short of entanglement [14]; BMV witnesses entanglement, so the negative prediction stands.) [SYN]

VI. MAGNITUDES

Decoherence. The Newtonian-limit scale is the Diósi–Penrose $\tau_G = \hbar/E_G$, with object-scale superpositions of 10^{10} – 10^{12} amu over second-to-hour holds. [STD/SYN]

Cosmological term — quantitatively excluded for Diósi–Penrose sourcing. Matching $\rho_\Lambda c^2 \approx 6 \times 10^{-10} \text{ J m}^{-3}$ over a Hubble time $t_H \approx 4 \times 10^{17} \text{ s}$ requires a mean injection rate density $\langle w \rangle \sim \rho_\Lambda c^2 / t_H \sim 1.5 \times 10^{-27} \text{ J m}^{-3} \text{ s}^{-1}$. The DP injection rate is fixed once the regularization length R_0 is:

$$\frac{dE}{dt} \sim \frac{3}{2} \frac{\hbar G m_N}{R_0^3}. \quad (7)$$

With the experimentally-allowed $R_0 \gtrsim 0.5\text{--}4 \text{ \AA}$ (Donadi [32]) and m_N the nucleon mass, $dE/dt \sim 10^{-41}\text{--}10^{-43} \text{ W}$ per nucleon; times the cosmic baryon number density $n_N \sim 0.25 \text{ m}^{-3}$, $w_{\text{DP}} \sim 10^{-42}\text{--}10^{-44} \text{ J m}^{-3} \text{ s}^{-1}$ — **12–16 orders of magnitude below $\langle w \rangle$** (an early-universe enhancement $\propto (1+z)^{3/2}$, $\sim 10^4\text{--}10^5$, does not close the gap). Reaching ρ_Λ would need $R_0 \sim 10^{-15} \text{ m}$, which Donadi excludes by ~ 5 orders.

Result 5 [SYN]. Diósi–Penrose-type ISP sourcing is not a viable dark-energy mechanism: the same regularization length R_0 bounded by the spontaneous-radiation experiments also fixes the cosmological injection, making the induced Λ many orders too small. The UG mechanism is real, but for DP-type sourcing its dark-energy realization is quantitatively excluded — a correct mechanism with an excluded magnitude.

Dark matter — no mechanism. Dark matter is a *content* problem (rotation curves, lensing, CMB third peak, structure formation, the Bullet Cluster’s lensing/gas offset). The construction adds no matter content; in the Newtonian limit it reproduces Newtonian gravity up to $R_0 \sim \text{\AA}$ [6], so no galactic-scale (MOND-like) modification; at linear order $\langle g(T[q]) \rangle = g(\langle T \rangle)$, so no extra mean attraction (only the fluctuations differ, sourcing decoherence not

mass). Its novel sector is bounded (Result 5) to $\gtrsim 12$ orders below ρ_Λ , far below $\rho_{\text{DM}} \sim 2\rho_\Lambda$, and of the wrong equation of state (heating $w > 0$, metric noise $w \approx 1/3$, vs. pressureless $w \approx 0$). The Bullet Cluster independently excludes modified gravity. [SYN/STD]

Result 6 [SYN]. The construction provides no dark-matter mechanism. With Result 5, ISP/RISP gravity addresses neither component of the dark sector — both are content / cosmological-gravity problems, while ISP reformulates dynamics.

VII. THE BINDING OBSTRUCTION: COVARIANCE OF THE BEABLE DYNAMICS

UG solves conservation but not covariance. Two features expose a preferred frame: (1) $\langle \eta_\nu \rangle$ is curl-free only in the frame where it carries no net momentum — the cosmic rest frame — so the well-definedness of $\Lambda(t)$ selects a foliation; (2) the beable is a configuration on a spatial slice, presupposing a notion of simultaneity. The status of this preferred frame:

universal foliation:	anisotropies $\sim \beta^2 \sim 10^{-6}$, excluded by Lorentz tests at $\sim 10^{-18}$,
confined to gravity/decoherence: evades bounds, signatures $\lesssim 10^{-7}$ (unobservable).	

(8)

So a preferred-frame RISP gravity is consistent only if the frame is confined to the non-unitary sector (**Route A**, empirically hidden). The covariant alternative (**Route B**) is more advanced than a free-particle construction: Tumulka realizes a Lorentz-invariant flash collapse not only for non-interacting particles [15] but for interacting *distinguishable* particles [16], inserting flashes into a Tomonaga-Schwinger evolution between arbitrary spacelike hypersurfaces — foliation-independent, no-signaling, reducing to non-relativistic GRW. The remaining gap is his stated limitation: the construction is not known for indistinguishable particles or variable particle number, i.e. for a *quantum field*, which is what sourcing gravity requires.

Any candidate must thread established no-go results: a relativistic *Markovian* collapse in standard QFT cannot keep both vacuum stability and no-signaling without nonstandard degrees of freedom; a Lorentz-covariant mass-coupled collapse cannot both avoid divergent energy production and forbid superluminal signaling [17]; and Gisin’s theorem [18] forbids

nonlinear (signaling) ensemble dynamics. These force abandoning one of {strict Lorentz invariance, Markovian dynamics, standard field ontology}. **The ISP beable abandons two** (non-Markovian and non-standard ontology), so it has more structural room than any Markovian collapse model. [OPEN]

Result 3 [SYN/OPEN]. The binding obstruction is not energy conservation (solved by UG) but covariance of the beable dynamics for a *quantum field*. A covariant interacting flash dynamics exists for *distinguishable* particles [16]; the field case required to source gravity is open. Being both non-Markovian and of non-standard ontology, the ISP beable forgoes two of the three properties the relativistic-collapse no-go theorems force one to abandon — giving it, in principle, the room to thread them.

VIII. FALSIFIABLE CONTENT AND HONEST SCOPE

Testable. (i) Gravitational decoherence with a non-exponential (Gaussian) onset at the DP scale, in matter-wave / levitated / optomechanical interferometry. (ii) The classical-vs-quantum-gravity fork via BMV: the construction predicts *no* gravitationally-induced entanglement, and a positive BMV result falsifies it (Result 4). (iii) A preferred-frame modulation of the decoherence rate at the $\lesssim 10^{-7}$ level (unobservable, but bounded from above by Lorentz tests if the frame is universal).

Not a prediction. The cosmological term is a real mechanism but not a dark-energy prediction (Result 5). The decoherence scale is Diósi–Penrose’s, not new; the contribution is the non-exponential shape (indivisibility) and the unified relativistic framing.

Known-address caveat. As with the program’s Bell-locality and frame-dependence analyses, the distinctive signatures (classical-gravity BMV behaviour, non-Markovian onset) are shared with other collapse / classical-gravity models. ISP provides a principled ontology and a unifying frame, not a unique fingerprint.

IX. OPEN PROBLEMS

1. Covariant interacting *field* beable dynamics — the binding problem [OPEN]. Construct a Lorentz-covariant, foliation-independent ISP beable dynamics for a quantum field that sources gravity. The distinguishable-particle case is solved [16]; the

field case is not. Any solution must thread the three no-go constraints above [17, 18]. Since the ISP beable already abandons two of the three, the ISP-specific question is whether its non-Markovianity provides enough room to build the covariant interacting field dynamics that Markovian collapse provably cannot. A positive answer makes relativistic ISP gravity consistent; a clean negative converts Route A into a theorem.

2. Curl-free violation beyond the mean [OPEN]. Characterize when the coarse-grained violation remains curl-free, and the back-reaction of $\delta\eta_\nu$.

3. Cosmological magnitude of Λ [RESOLVED, negative]. Result 5: DP-type sourcing with the allowed R_0 gives Λ 12–16 orders too small.

4. BMV entanglement [RESOLVED for this construction]. Result 4: classical channel, no entanglement [11]; the 2025 dispute is a model-definition issue.

X. INVESTIGATING THE BINDING OBSTRUCTION: RELATIVISTIC STOCHASTIC FIELD THEORY AND A FREE-FIELD TESTBED

A. The reduction

Stripped of gravity, open problem 1 is: construct a Lorentz-covariant, real-time stochastic process, with genuine non-negative probabilities, over field configurations, reconstructing the unitary QFT — relativistic stochastic mechanics for fields done with honest probabilities (the field-theoretic Nelson/Barandes problem). The beable is a configuration φ on a slice (foliation enters); indivisibility is the candidate escape from the no-go.

B. The covariant structure exists — in Euclidean signature

A relativistic QFT carries a genuine, manifestly covariant probability measure over field configurations: the Euclidean measure e^{-S_E} (Euclidean rotation = Lorentz after Wick rotation), with Osterwalder–Schrader reconstruction recovering the Lorentzian theory via reflection positivity. Guerra and Ruggiero proved the Euclidean–Markov field is the ground-state stochastic process of Nelson’s stochastic mechanics [19, 20]. The Markov structure is diagnostic: the *free* field is Markov, the *interacting* field generically non-Markov. So non-Markovianity is the generic structure of interacting relativistic random fields. [STD]

C. The obstruction (refined): see Result 10

The Euclidean measure is imaginary-time; the real-time path integral has oscillatory e^{iS} weights. A stochastic *process* lives at the time-sliced, intermediate level, which is not manifestly covariant even though the final QFT is. We sharpen and partly correct this framing in Sec. XH: the ISP transition probabilities are positive, so the binding obstruction is covariance, not positivity. [STD framing]

D. Why indivisibility is the candidate tool

In Barandes' construction the quantum phase is encoded in the non-Markovian memory, not in complex weights [1, 3]. Nelson's stochastic mechanics is the Markovian-diffusion *special case* of the indivisible class; its Markov-diffusion structure selects a time direction and resists covariantization (the fifty-year obstruction, studied even for the free scalar field [29]). The ISP-specific question: does the general indivisible class contain a Lorentz-covariant interacting-field member? Three facts make it well-posed: interacting Euclidean fields are already non-Markov; the no-go is for Markovian dynamics; it localizes the obstruction to one object. *Caveat (Sec. XE): the free-field test shows non-Markovianity is not what buys covariance per se.* [SYN]

E. The free-scalar-field testbed: a worked result

We carried out the test. **Outcome: a covariant genuine-probability ISP exists for the free field — but through Gaussianity, not non-Markovianity.** The free field is one oscillator per mode k with $\omega_k = \sqrt{k^2 + m^2}$; Nelson's process makes each an independent Ornstein–Uhlenbeck (Markov) process with correlation

$$\langle \tilde{\varphi}(k, t) \tilde{\varphi}(-k, t') \rangle = \frac{1}{2\omega_k} e^{-\omega_k |t-t'|}, \quad (9)$$

exactly the Euclidean propagator in the mixed (k, τ) representation [20]. A covariant genuine-probability description exists two ways: (A) the Euclidean random field e^{-S_E} is a genuine $O(4)$ -invariant probability measure on 4D configurations; (B) the Gaussian vacuum has a positive Wigner functional [21], a genuine phase-space probability under classical Liouville flow. So relativistic ISP exists for the free field. But (1) it works via Gaussianity,

not non-Markovianity (the process is Markov; no memory kernel needed), disconfirming the Sec. 10.4 reading; and (2) by Hudson’s theorem a pure state has a non-negative Wigner function *iff* it is Gaussian, so any interaction makes route (B) negative — the sign problem as a theorem — while route (A)’s $|\Psi[\varphi]|^2$ stays positive but the interacting Euclidean field is non-Markov and its covariant real-time process is unestablished.

Result 7 [SYN]. Relativistic ISP exists for the free scalar field (Gaussian, integrable: positive Wigner [21] / Euclidean–Markov field [20]) — but for the trivial reason and *without* non-Markovianity, so it neither advances nor evidences the interacting conjecture. Hudson’s theorem makes the interacting obstruction precise: positive phase-space probability is Gaussian-only.

F. Beyond the free field: the case map and the reduction to pure foliation-independence

Arrange the cases by (free vs interacting) $\times$ (non-relativistic QM vs relativistic field). Three of four cells are settled: *interacting QM is already an ISP* (Barandes’ theorem builds a config-space indivisible process for any unitary system, including Wigner-negative states — so Hudson is a *phase-space* obstruction that config-space ISP evades); the *covariant interacting measure exists* (constructive QFT: $P(\phi)_2$, ϕ_2^4 Euclidean measures [22]); and *2D dualities* (Thirring / sine-Gordon via bosonization [23], Ising via free fermions) give covariant interacting ISPs through free duals. Formally, Barandes’ construction applied to the interacting QFT Hilbert space and $U(t)$ already yields a config-space ISP *in a frame*; the residue is purely *foliation-independence*.

Result 8 [SYN]. The candidate obstructions fall one by one (Wigner-negativity is phase-space; the covariant interacting measure exists; 2D dualities give covariant interacting ISPs). The binding obstruction narrows to foliation-independence of the frame-dependently-existing interacting field process — modulo the rigorous QFT extension of Barandes against Haag’s theorem [24].

G. Does Haag’s theorem obstruct, or merely complicate?

Haag forbids only the interaction picture; it does not forbid the interacting theory, which exists Poincaré-covariantly in its own representation (Wightman; constructive QFT [22] builds it by abandoning the equal-time-CCR / interaction-picture assumption). The configuration-space representation ISP needs exists for interacting fields — the Schrödinger functional representation $(\Psi[\varphi], U(t) = e^{-iHt}, \text{full } H)$ — which is renormalizable [25] and Haag-evading, so a *frame-dependent* interacting-field ISP exists. But the Schrödinger functional is *not manifestly Lorentz invariant*, and the manifestly-covariant route is the Tomonaga–Schwinger (interaction-picture) evolution — the tool that made the distinguishable-particle case covariant — which Haag obstructs for fields.

Result 9 [SYN]. Haag’s theorem does *not* obstruct the existence of an interacting-field ISP (the Schrödinger functional representation [25] supplies a frame-dependent one); it *does* obstruct the manifestly-covariant Tomonaga–Schwinger route, explaining why the distinguishable-particle success [16] does not transfer to fields. The residue is sharp and non-Haag: promote the non-manifest Lorentz covariance of the Schrödinger-functional interacting theory to a manifestly-covariant config-space ISP without the interaction picture. A prior covariant beable-QFT attempt — Nikolić’s many-fingered-time Bohmian QFT [26] — *claims* exactly this, at a formal level whose rigorous status against Haag’s theorem is itself the open question.

H. Attempting the residue: a four-way no-free-lunch

We attempted the residue via the most tractable (Euclidean / OS) route; the honest outcome is a structured obstruction. Four desiderata: **(C)** manifest covariance, **(R)** real time, **(P)** genuine probability, **(E)** a beable evolving in physical time. The natural constructions

each achieve three and drop one:

route	C	R	P	E	drops
Barandes $ U ^2$	—	✓	✓	✓	manifest covariance
Euclidean / OS	✓	—	✓	—	real-time process
stochastic quantization	✓	—	✓	✓	physical-time evolution
complex Langevin	✓	✓	—	✓	positivity (sign problem)

(10)

The Barandes transition probabilities $|\langle q'|U(t)|q\rangle|^2$ are genuinely non-negative (no sign problem at the pairwise level; the quantum-ness is in the non-Markovian incompatibility of the multi-time joints) — which corrects the “process-level sign problem” language of Sec. 10.3: the obstruction for ISP is covariance, not positivity. OS reconstruction yields a Hilbert space and correlators, not a real-time process. Stochastic quantization (Parisi–Wu) gives a covariant genuine process but in fictitious Langevin time (a 4D block-universe beable). Complex Langevin restores real time but with complex weights. The deep tension: manifest covariance pulls toward a 4D block-universe object, while ISP wants a beable evolving in physical time.

Result 10 [SYN/OPEN]. The Euclidean-reconstruction attempt does not close the residue; with its siblings it reveals a four-way no-free-lunch among {manifest covariance, real time, genuine probability, physical-time-evolving beable} — every known construction drops exactly one. A manifestly covariant relativistic ISP appears to require either a hidden preferred frame (Route A) or the abandonment of physical-time evolution for a 4D-history beable. This is a structured map of the obstruction, not a no-go: no theorem forbids a fifth route, but none of the standard four delivers all four desiderata.

I. Working the residue: the many-fingered-time route and the foliation-relative-beable condition

A direct attack on the residue identifies a fifth route the Result-10 table omits, and the precise condition under which it threads the no-free-lunch. *The fifth route is many-fingered time:* evolve a Schrödinger-functional state $\psi[\Sigma, \varphi]$ over all spacelike hypersurfaces Σ by the Tomonaga–Schwinger equation with the *full* interacting Hamiltonian density (not the interaction picture). This is manifestly covariant (no preferred Σ), real-time,

config-space-positive ($|\psi[\Sigma, \varphi]|^2$), carries a beable on each slice — formally all four desiderata — and *evades Haag* (full H in the Schrödinger-functional representation [25], not the free $\leftrightarrow$ interacting intertwiner); it is what Nikolić’s covariant Bohmian QFT [26] targets. *The wall* is foliation-independent equivariance: requiring the Born distribution $|\psi[\Sigma]|^2$ on *every* Σ over-determines a single 4D beable history $\varphi(x)$ (marginals on crossing slices cannot all be Born except in free/Gaussian cases) — the standard obstruction to covariant Bohmian QFT. *The ISP-specific lever*: indivisibility commits only to single-slice Born marginals plus a non-divisible within-foliation transition structure; it does not posit a consistent trans-foliation joint history, which is what the over-determination constrains — so an indivisible process can carry Born marginals on every Σ where a Bohmian single trajectory cannot. *The enabling condition* is a *foliation-relative beable*: configs defined per slice, with no foliation-independent point-value $\varphi(x)$. If the beable is instead a 4D field with point-values, the over-determination bites and the route collapses to the block-universe (Euclidean) option. The foliation-relative reading is arguably relativistically natural (no absolute simultaneity $\Rightarrow$ no absolute “configuration now”), but its coherence is the new open question.

Result 11 [SYN/OPEN]. A fifth route exists that the Result-10 table omits: a many-fingered-time, Schrödinger-functional, *indivisible* process. It evades Haag (full H) and would thread the four-way no-free-lunch *provided* a foliation-relative beable is a coherent ontology — exactly the condition under which indivisibility evades the equivariance over-determination that sinks covariant Bohmian QFT. The residue thus sharpens to one ISP-specific open question: *are foliation-relative indivisible beables coherent?* Two technical issues remain even then: Tomonaga–Schwinger integrability against Schwinger-term anomalies in $[\mathcal{H}(x), \mathcal{H}(y)]$, and a rigorous construction. A structured advance, not a solution; every step is conjectural and the ontology may prove incoherent.

J. Is the foliation-relative beable coherent? A positive case

The verdict on the question Result 11 isolated is a qualified *yes*: the technical incoherence-candidates dissolve, and the residue is a single defensible interpretive commitment, not a contradiction. *Established precedent*: Aharonov and Albert [27] proved relativistic quantum states are genuinely hypersurface-relative — the covariance resides exclusively in the experimental probabilities, not the states — and relativistic Bohmian mechanics admits a

class of empirically equivalent probability spaces, one per foliation [28]; the foliation-relative indivisible beable is the beable-ontology analogue. *The beable is $\varphi(x, n)$* , a field value at each point relative to a local timelike direction n (the slice normal); unlike single-valued $\varphi(x)$ it is not over-determined, since different foliations read off different n -components. *Microcausality dissolves the incoherence candidates*: on a shared region R the Born marginal is fixed by the local algebra $\mathcal{A}(D(R))$, identical for crossing slices; records/outcomes are localized, hence foliation-independent, so observers agree; the joint of two spacelike outcomes is fixed by $\mathcal{A}(A) \vee \mathcal{A}(B)$, foliation-independent and equal to QM's. Foliation-relativity is confined to the unobservable global structure — exactly as relativity confines frame-relativity to simultaneity. *Indivisibility is decisive*: a deterministic Bohmian trajectory must dynamically preserve equivariance ($|\psi|^2$) on each foliation (clean only on a preferred one, forcing Bohmian toward a preferred frame), whereas the indivisible process merely posits the Born marginal on each slice, with no preservation burden.

Result 12 [SYN/OPEN]. Foliation-relative indivisible beables are *tentatively coherent*. Every technical incoherence-candidate (crossing-slice consistency, record/observer agreement, Bell correlations) dissolves via microcausality: local facts are foliation-independent, only the global beable structure is foliation-relative — the structure Aharonov–Albert established for relativistic states [27] and relativistic Bohmian mechanics realizes as a per-foliation class [28]. Indivisibility removes the equivariance-preservation burden that forces Bohmian toward a preferred foliation. What remains is not a contradiction but one explicit interpretive commitment — that the beable (reality), not merely the description, can be frame-relative — coherent by parallel to relativity's frame-relative simultaneity, but a genuine ontological postulate. Caveats: coherence by dissolution-of-objections plus parallels, not a theorem; the leap from frame-relative state to frame-relative beable is substantive; and ontological coherence is separate from existence of the construction (Result 11).

K. The technical move: Tomonaga–Schwinger integrability and the Schwinger anomaly

The remaining outright-killer is whether the full- H Tomonaga–Schwinger evolution $i\delta\psi[\Sigma]/\delta\sigma(x) = \mathcal{H}(x)\psi[\Sigma]$ is integrable (path-independent in hypersurface space) for an interacting field. The verdict is favorable in the semiclassical setting. *Distinct-point inte-*

grability holds by microcausality: the renormalized $\mathcal{H}(x) = T^{00}(x)$ is local, so renormalized microcausality gives $[\mathcal{H}(x), \mathcal{H}(y)] = 0$ for $x \neq y$ spacelike, for the *full* interacting density. *The coincidence anomaly is a phase that cancels in the beable:* the deformation algebra carries Schwinger terms at $x = y$ — the classical part ($\propto T^{0k} \partial_k \delta$, the tangential generator) is the correct hypersurface-deformation algebra, faithfully represented, while the anomalous part is a c-number central extension (the conformal/Schwinger anomaly). For Hermitian $\mathcal{H}$, $[\mathcal{H}(x), \mathcal{H}(y)]$ is anti-Hermitian, so its c-number part is purely imaginary (a phase, not a norm change), and being a c-number it is field-configuration-independent (a global phase $e^{i\alpha[\Sigma]}$). Either way it cancels in the Born-marginal beable $|\psi[\Sigma, \varphi]|^2$: the state’s phase may be anomalous, but the beable distribution is path-independent. Indivisibility helps again — the beable being a modulus is exactly what makes it immune to the phase anomaly.

Result 13 [SYN/OPEN]. In the semiclassical setting the Tomonaga–Schwinger integrability worry does *not* kill the fifth route. Distinct-point integrability holds by renormalized microcausality ($[\mathcal{H}(x), \mathcal{H}(y)] = 0$, $x \neq y$); the coincidence Schwinger anomaly is an anti-Hermitian c-number — a global phase cocycle on $\psi[\Sigma]$ — which cancels in the Born-marginal beable. So the beable is integrable even when the state’s phase is anomalous. Holds provided the $[\mathcal{H}, \mathcal{H}]$ anomaly is c-number (no anomalous operator terms beyond the classical deformation algebra; true for standard renormalizable QFT) and in the semiclassical setting. Defusing this obstruction is necessary, not sufficient: a rigorous construction gluing the per-foliation Barandes ISPs into one covariant object with consistent all-foliation Born marginals remains undone, and a subtler obstruction is not excluded.

L. Honest status

This converts open problem 1 from an aspiration into a well-posed problem with a fifty-year literature, a known covariant structure (Euclidean–Markov), and a named obstruction. The free-field testbed settles the easy case — relativistic ISP exists — but through Gaussianity, not non-Markovianity. Every other candidate obstruction (conservation, Wigner-negativity, measure-existence, interacting-existence, Haag-existence) has been ruled out, localizing the residue to a manifestly-covariant, non-interaction-picture construction of the frame-dependently-existing interacting-field config-space ISP. No one — Nelson, Guerra–Ruggiero, or Barandes — has produced one. Realistically a decade-scale problem in mathe-

matical physics, now stated correctly. [OPEN]

XI. EPISTEMIC STATUS AND ROUTE TO ACCEPTANCE

This paper, like the program, offers RISP *no unique measurable prediction*: the relativistic sections reproduce the observables of QFT+GR, the one measurable deviation (the gravitational decoherence of the companion decoherence paper) is *non-relativistic* ISP shared with the collapse-model family, and the genuinely relativistic content (the foliation-relative beable) is, by Result 12 and the unobservability argument, observationally hidden. It is worth stating plainly *why* the relativistic construction is worth pursuing and *how* such a theory could ever be accepted.

a. The value is ontological unification, not unique prediction. Ontologies are routinely adopted for explanatory scope and unification rather than by crucial experiments — atomism before atoms were resolved, the reality of fields, of spacetime curvature. RISP’s claim is of that kind: the coherent relativistic ontology that a confirmed ISP would generalize to. Its route to acceptance is inference-to-the-best-explanation, not a RISP-specific test — a legitimate but non-empirical path, which we name as such.

b. Three conditions, one of them binding.

1. **ISP must be empirically confirmed** — the optional non-unitary channel must exist, and the two-axis test must land in the ISP cell. Even then, what is confirmed is the *class* “non-Markovian gravitational objective-collapse”; selecting Barandes’ ISP within it is itself a coherence judgement.
2. **RISP must exist as a constructed, coherent theory — the binding condition.** One cannot generalize a confirmed ISP to RISP if the covariant interacting beable dynamics (Sec. X) is not built. This paper cleared that construction’s would-be-killers (Results 11–13) but did not build it. Far from lowering the burden, the acceptance route *rests* on it: no construction, no ontology to generalize to.
3. **RISP must win the ontology competition** — against other relativistic completions (Tumulka’s relativistic flash GRW, Bohmian QFT, the spacetime-from-entanglement program), on coherence and naturalness, since these are roughly empirically equivalent. RISP’s case is that indivisibility is the *confirmed* ISP feature and

the foliation-relative beable its natural relativistic completion — an argument to be made, not an automatic victory.

c. The historical caveat. Confirmation of a *dynamics* underdetermines its *ontology*. Quantum mechanics has been confirmed for a century and its ontology remains contested (Copenhagen, Bohm, Everett, collapse, ISP all coexist); the luminiferous aether was the “natural” ontology for confirmed wave optics and was wrong. So even a confirmed ISP may leave the relativistic ontology as unsettled as QM’s interpretation is today; predicting that RISP *would* be widely adopted is itself an overclaim.

d. Program priorities. (i) Get ISP confirmed (the two-axis decoherence test); (ii) *build* RISP (complete the construction of Sec. X — the linchpin); (iii) argue RISP’s coherence against rival relativistic ontologies. If all three hold, RISP could become the accepted QFT+GR ontology by the unification route — and it should be presented as inference-to-the-best-explanation, named as such, never as a crucial-experiment result. [**meta** / **epistemic**]

XII. CONCLUSION

Non-conserved matter — of which the indivisible beable is one instance — can source classical gravity relativistically once one adopts unimodular gravity: the energy–momentum non-conservation that ordinarily obstructs the coupling is absorbed into a dynamical cosmological term, splitting into a mean part (dark-energy-like) and a fluctuating part (gravitational decoherence). This unimodular sourcing is Markovian-agnostic; making the *source itself* a genuinely relativistic indivisible beable is the separate open problem investigated in Sec. X. The framework reduces, in the Newtonian limit, to Diósi–Penrose / Tilloy–Diósi physics and to a falsifiable Gaussian-onset decoherence. Two quantitative questions are settled: the construction is a classical channel, predicting a negative BMV result (falsified by a positive one); and the cosmological term, while a genuine consequence of the unimodular mechanism, is quantitatively excluded as dark energy for Diósi–Penrose-type sourcing. The binding open problem is the covariance of the beable dynamics for a quantum field, which we dissect to a four-way no-free-lunch and a single manifestly-covariant non-interaction-picture construction.

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